

Geethanjali College of Engineering and Technology

(AUTONOMOUS)

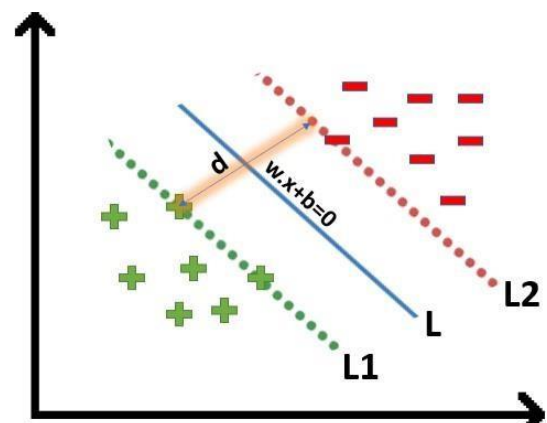
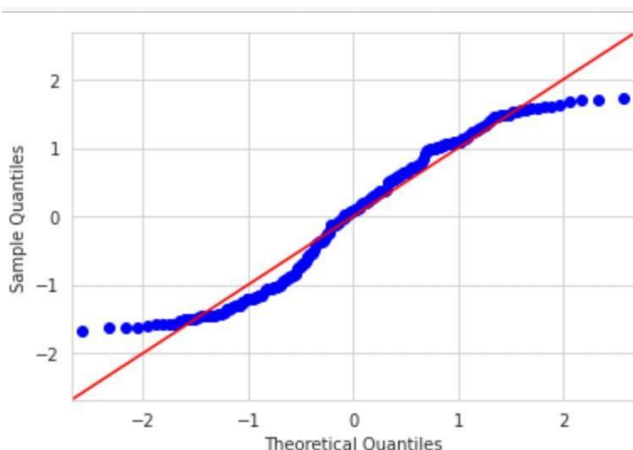
(Accredited by NAAC "A+" Grade; CSE ,EEE,ECE B.Tech

Program Accredited by NBA; Approved by AICTE, New Delhi, Affiliated to JNTUH)

Cheeryal (V), Keesara (M), Medchal (Dist),Telangana State.

STATISTICS FOR MACHINE LEARNING LABORATORY WORKBOOK

III Year B. Tech-I Semester (AY 2026-27)



Name: Roll No:

Branch: Section:

Geethanjali College of Engineering and Technology

AUTONOMOUS

(Accredited by NBA, NAAC “A+” Grade)

Cheeryal (V), Keesara (M), Medchal District – 501 301 (T S)



Geethanjali

STATISTICS FOR MACHINE LEARNING LAB

Subject Code – (20MA31L01)

LABORATORY MANUAL

III Year B.Tech I Semester

COMPUTER SCIENCE AND ENGINEERING - ARTIFICIAL INTELLIGENCE AND MACHINE

LEARNING - CSE(AI&ML)

2026-2027

DEPARTMENT OF FRESHMAN ENGINEERING

Geethanjali College of Engineering and Technology
Geethanjali College of Engineering and Technology
AUTONOMOUS
(Accredited by NBA, NAAC “A+” Grade)
Cheeryal (V), Keesara (M), Medchal District – 501 301 (T S)



CERTIFICATE

This is to certify that Mr. / Miss _____
*has satisfactorily completed _____ number of programmes in the **Statistics for Machine Learning lab.***

Roll No: _____

Branch: _____

Year: _____

Academic Year: _____

Head of the Department

Faculty In-charge

Internal Examiner

External Examiner

Geethanjali College of Engineering and Technology
AUTONOMOUS

Subject code: 20MA31L01

L	T	P/D	C
-	-	2/-	1

Pre-requisite(s): 20MA31001- Statistics for Machine Learning

Course Outcomes:

At the end of the course, student would be able to use R software to:

- CO1: Construct frequency distribution for the given data.
- CO2: Obtain specified set of samples from large data through random variables using Acceptance-Rejection (AR) algorithm.
- CO3: Estimate and test the significance of the specified population parameter(s) for the given data.
- CO4: Reduce the dimensionality of large data sets using Principal Component Analysis (PCA), Logistic Regression (LR), K-means clustering and Support Vector Machine (SVM) for selecting variables.

List of Experiments:

1. Introduction to R Programming
2. Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)
3. Acceptance/Rejection Sampling in R
4. Maximum likelihood and method of moments estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R
5. Testing of hypothesis based on t-test, chi-square test and F-test, Confidence interval estimation in R
6. Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi-collinearity in R
7. Logistic Regression in R
8. Classification with SVM in R
9. Dimensionality reduction with PCA in R and K-means Clustering in R
10. Kernel PCA and Kernel SVM in R

List of Additional programs

1. Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R Programming
2. Simple Linear Regression :Testing overall hypothesis and testing significance of variables
3. Multiple linear regression: Testing overall hypothesis and testing significance of individual variables, model selections and prediction in R programming.

Vision of the Institute:

Geethanjali visualizes dissemination of knowledge and skills to students, who would eventually contribute to the well-being of the people of the nation and global community.

Mission of the Institute:

- To impart adequate fundamental knowledge in all basic sciences and engineering, technical and Interpersonal skills to students.
- To bring out creativity in students that would promote innovation, research and entrepreneurship.
- To Preserve and promote cultural heritage, humanistic and spiritual values promoting peace and harmony in society.

Vision of the Department:

Department of AI&ML

To produce globally competent and socially responsible computer science engineers contributing to the advancement of engineering and technology which involves creativity and innovation by providing excellent learning environment with world class facilities.

Mission of the Department:

- To be a centre of excellence in instruction, innovation in research and scholarship, and service to the stake holders, the profession, and the public.
- To prepare graduates to enter a rapidly changing field as a competent computer science engineer.
- To prepare graduate capable in all phases of software development, possess a firm understanding of hardware technologies, have the strong mathematical background necessary for scientific computing, and be sufficiently well versed in general theory to allow growth within the discipline as it advances.
- To prepare graduates to assume leadership roles by possessing good communication skills, the ability to work effectively as team members, and an appreciation for their social and ethical responsibility in a global setting

Program Educational Objectives (PEO's):

Program Educational Objectives (PEOs) are broad statements that describe what graduates are expected to attain within a few years of graduation. The PEOs for Computer Science and Engineering (Artificial Intelligence and Machine Learning) graduates are:

PEO1 To provide graduates with a good foundation in mathematics, sciences and engineering fundamentals required to solve engineering problems that will facilitate them to find employment in industry and / or to pursue postgraduate studies with an appreciation for lifelong learning.

PEO 2: To provide graduates with analytical and problem solving skills to design algorithms, other hardware / software systems, and inculcate professional ethics, inter-personal skills to work in a multi-cultural team.

PEO 3: To facilitate graduates get familiarized with state of the art software / hardware tools, imbibing creativity and Innovation that would enable them to develop cutting-edge technologies of multi-disciplinary nature for societal development.

Program Outcomes (POs):

1. Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

2. Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

3. Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

4. Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

5. Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

6. The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

8. Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

10. Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11. Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Specific Outcomes (PSOs):

PSO 1: Demonstrate competency in Programming and problem-solving skills and apply those skills in solving computing problems.

PSO2: Select appropriate programming languages, Data structures and algorithms in combination with modern technologies and apply them in developing innovative solutions.

PSO 3: Demonstrate adequate knowledge in the concepts and techniques of artificial intelligence and machine learning, apply them in developing intelligent systems to solve real world problems

Mapping of Lab Course outcomes with Program outcomes:

CO-PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PS O1	PS O2	PS O3
CO1	3	2			-	-	-	-	-	2	-	2	-	-
CO2	3	2			-	-	-	-	-	2	-	2	-	-
CO3	3	2		3	-	-	-	-	-	3	-	3	-	-
CO4	3	2	3	3	-	-	-	-	-	3	-	3	-	-

-	No Correlation	1	Slight (low)	2	Moderate (medium)	3	Substantial (High)
---	----------------	---	--------------	---	-------------------	---	--------------------

Prerequisites:

Statistics for Machine Learning (20MA31001)

INSTRUCTIONS TO THE STUDENTS:

1. Students are required to attend all labs.
2. Students should be dressed in formals when attending the laboratory sessions.
3. Students will work individually in computer laboratories.
4. While coming to the lab bring the observation book and Work book etc.
5. Before coming to the lab, prepare the pre-lab questions. Read through the lab experiment to familiarize you.
6. Utilize 3 hours' time properly to perform the experiment and noting down the outputs.
7. If the experiment is not completed in the prescribed time, the pending work has to be done in the leisure hour or extended hours.
8. You will be expected to submit the completed work book according to the deadlines set up by your instructor.

INSTRUCTIONS TO LABORATORY TEACHERS:

1. Observation book and lab records submitted for the lab work are to be checked and signed before the next lab session.
2. Students should be instructed to switch ON the power supply after the connections are checked by the lab assistant / teacher.
3. The promptness of submission should be strictly insisted by awarding the marks accordingly.
4. Ask Viva Voce questions at the end of the experiment.
5. Do not allow students who come late to the lab class.

6. Encourage the students to do the experiments innovatively.
7. Fill Continuous Evaluation Sheet, on regular basis.
8. Ensure that the students are dressed in formals

Evaluation pattern:

Total Lab Marks –100 (Internal - 40 Marks & External – 60 Marks)

Internal Assessment:

1. Continuous valuation is done for 10 Marks based on:
 - a. Submission of workbook in time (5 Marks)
 - b. Performance of Program (5 Marks)
2. During the semester, two Lab Internal examinations are conducted for 10 Marks and average will be considered.
3. Viva-voce carries 10 Marks.
4. Project Based Learning (PBL) presentation carries 10 marks.

External Assessment: At the end of semester, external lab exam is conducted for 60 Marks with one external examiner and one internal examiner as per regulation.

- Write up and program code (25 Marks)
- Execution of Program (25 Marks)
- Viva-Voce (10 Marks)

PERFORMANCE INDICATOR

S No.	Name of the Program	Page No.	Date of Programme	Date of Submission	Marks awarded	Faculty Sign.	Remarks
1	Introduction to R Programming						
2	Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)						
3	Acceptance/Rejection Sampling in R						
4	Maximum likelihood and method of moments estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R						
5	Testing of hypothesis based on t-test, chi-square test and F-test. Confidence interval estimation in R						
6	Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi-collinearity in R						
7	Logistic Regression in R						
8	Classification with SVM in R						
9	Dimensionality reduction with PCA in R and K-means Clustering in R						
10	Kernel SVM in R						

ADDITIONAL PROGRAMMES							
S No.	Name of the Program	Page No.	Date of Programme	Date of Submission	Marks awarded	Faculty Sign.	Remarks
1	Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R programming.						
2	Simple Linear Regression in R programming.						
3	Multiple linear regression: Testing over all hypothesis and testing significance of individual variables, model selections and prediction in R programming.						

S No.	INDEX	Page No.
1	Introduction to R Programming	
2	Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)	
3	Acceptance/Rejection Sampling in R	
4	Maximum likelihood and method of moments estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R	
5	Testing of hypothesis based on t-test, chi-square test and F-test. Confidence interval estimation in R	
6	Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi-collinearity in R	
7	Logistic Regression in R	
8	Classification with SVM in R	
9	Dimensionality reduction with PCA in R and K-means Clustering in R	
10	Kernel SVM in R	
	Additional Programmes	
1	Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R programming	
2	Simple Linear Regression :Testing overall hypothesis and testing significance of variables	
3	Multiple linear regression: Testing overall hypothesis and testing significance of individual variables, model selections and prediction in R programming	

1. Introduction to R Programming

R is a programming language and free software environment for statistical computing and graphics that is supported by the R Foundation for Statistical Computing. The R language is widely used among statisticians and data miners for developing statistical software and data analysis.

R is a programming language and software environment for statistical analysis, graphics representation and reporting.

The following are the important features of R

- R is a well-developed, simple and effective programming language which includes conditionals, loops; user defined recursive functions and input and output facilities.
- R has an effective data handling and storage facility,
- R provides a suite of operators for calculations on arrays, lists, vectors and matrices.
- R provides a large, coherent and integrated collection of tools for data analysis.
- R provides graphical facilities for data analysis and display either directly at the computer or printing at the papers.

To Install R and R Packages:

1. Open an internet browser and go to www.r-project.org.
2. Click the "download R" link in the middle of the page under "Getting Started."
3. Select a CRAN location (a mirror site) and click the corresponding link.
4. Click on the "Download R for WINDOWS" link at the top of the page.
5. Click on the file containing the latest version of R under "Files."
6. Save the **.pkg** file, double-click it to open, and follow the installation instructions.
7. Now that R is installed, you need to download and install R Studio.

To Install R Studio:

1. Go to www.rstudio.com and click on the "Download R Studio" button.
2. Click on "Download R Studio Desktop."
3. Click on the version recommended for your system, or the latest Mac version, save the **.dmg** file on your computer, double-click it to open, and then drag and drop it to your applications folder.

Install R Packages:

1. Download and install a package (you only need to do this once).
2. To use the package, invoke the library (package) command to load it into the current session. (You need to do this once in each session, unless you customize your environment to automatically load it each time).

Installing and Loading Packages:

R to load the package's library in R session

```
> library("MASS") #R now knows all the functions that are scanned in the MASS package
```

1.1. Basics and R as a Calculator:

- i. 31 * 78 # Multiplication

```
[1] 2418
```

- ii. 697/41 #division

```
[1] 17
```

- iii. Assign the value of 39 to x

```
x<-39
```

```
> x
```

```
[1] 39
```

- iv. Assign the value of 22 to y

```
> y<-22
```

```
> y
```

```
[1] 22
```

- v. Make z the value of x - y

```
> z<- x - y
```

- vi. Display the value of z in the console

```
> z
```

```
[1] 17
```

- vii. Find 10 random numbers between 1 and 100. (Hint: you can use sample() function)

```
> sample(1:100,10)
```

```
[1] 77 74 89 23 39 84 63 12 91 16
```

1 Example: Find the value of 21 is multiplied by 23, 32 is multiplied by 62; 1256 divided by 2;
Assign the value of 26 to x variable, 33 to y variable and hence find the value of z such that $z = x+y$; $x-y$;

R Code with Output:

2 Example: Find 5 random numbers between 1 to 100 using sample function

R Code with Output:

1.2. Creating Vectors:

A vector is a basic data structure of R containing the same type of data.

R Code for vector calculations:

x = c(observations / data) # Store the observations in x

x+y # Addition of two vectors

x-y Subtraction of two vectors

x*y Multiplication of two vectors

1. Example: The numbers below are the first ten days of rainfall amounts in 1996. Read them into a vector using the vector function named as “rainfall”

0.1, 0.6, 33.8, 1.9, 9.6, 4.3, 33.7, 0.3, 0.0, 0.1

R-Code with Output:

2. Example: Find the addition, subtraction and multiplication of two vectors

A= 20,22,24,25,21,26,25,26,21,25,26,24,25,25,26,25,24,25,26,25

B= 15,14,15,16,15,14,18,16,14,15,15,17,15,15,14,15,16,17,18,19

R-Code with Output:

1.3. Creating Matrices:

To create matrices we will use the `matrix()` function. The `matrix()` function takes the following arguments:

- data an R object (this could be a vector).
- nrow the desired number of rows.
- ncol the desired number of columns.
- byrow a logical statement to populate the matrix by either row or by column

R Code for Matrix Calculations:

```
A=matrix(c(data),nrow=no.of rows, ncol=no.ofcolumns, byrow=TRUE)
                                     # Store the data in matrix A
A+B, # Addition,
A-B, # Subtraction
A%*%B # Multiplication of two matrices,
t(A) # Transpose of a matrix
rankMatrix(A) # Rank of a matrix
eigen(A) # Eigen values and eigen vectors
```

1. Example: If x is a matrix specified by the R command `x = matrix(nrow=3, ncol=3, data=c(1,8,2,1,5,4,3,9,6), byrow=TRUE)`, then the output of the R command `x%*%t(x)` is?

R-Code with Output:

2. Example: If $A = \begin{bmatrix} 0 & 3 & 1 \\ 5 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ are two matrices then check whether the two matrices satisfy commutative property or not. And also find transpose of each matrix using R program.

a) R-Code with Output:

b) R-Code with Output:

3. **Example:** Find the rank of a matrix $A = \begin{bmatrix} 3 & 2 & 4 & 0 & -4 \\ 2 & 1 & 2 & 1 & 5 \\ 0 & 0 & 3 & -2 & 1 \\ 4 & 5 & 1 & 1 & 2 \\ 2 & 1 & 5 & -1 & 0 \\ 1 & 2 & 2 & 2 & 1 \end{bmatrix}$

R-Code with Output:

4. **Example:** Find the eigen values and eigen vectors of a given matrix $A = \begin{bmatrix} 1 & 8 & 9 & 16 \\ 2 & 7 & 10 & 15 \\ 3 & 6 & 11 & 14 \\ 4 & 5 & 12 & 13 \end{bmatrix}$

R-Code with Output:

1.4. Sequence and drawing a Scatterplot

R Code:

```
seq(from = .., to = .., by = ...)
plot(x, y, main="Title", xlab="x", ylab="y")
abline(lm(x~ y), col = ".....")
```

1. Example: Print Numbers from 0 to 5 by the difference 0.5

R-Code with Output:

2. Example: carage :5,7,8,7,2,2,9,4,11,12,9,6

Carspeed :99,86,87,88,111,103,87,94,78,77,85,86

Show these two variables in scatter plot and identify is there any relationship between these two variables.

R-Code with Output:

Viva Questions:

1. Write the syntax for installing package MASS?

Ans.

2. Write the syntax to load MASS package in library?

Ans.

3. Write the R code to find the value of $Z = X + Y$. where $X=26$, $Y = -10$.

Ans.

2. Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)

Absolute Frequency, Relative Frequency:

The number of observations in a particular class / category is called the **Absolute Frequency** or **Frequency**.

The number of observations in a particular class / category is divided by total frequency is called **Relative Frequency**.

2.1. Absolute and Relative Frequency.

R Code:

```
table(data vector)           #creates the absolute frequency of the data vector of the given
data
```

Enter data as **x**

```
table(x)                     # displays Absolute Frequencies
```

```
table(x)/length(x)          # displays Relative Frequencies
```

1. Example: Results of 10 persons declared in two categories as Pass (P) and Fail (F) is categorized as 1 and 2 respectively. Find Absolute and Relative Frequencies.

P, F, P, F, F, P, P, F, P, P

1, 2, 1, 2, 2, 1, 1, 2, 1, 1

R Code with Output:

2. Example: The time (in minutes) taken to complete the puzzle was recorded as follows and stored in a data vector as puzzle time. Find Absolute and Relative frequencies.

8,13,21,9,15,9,21,13,8,8,13,15,8,21,9,15,8,13,21

R Code with Output:

2.2. Frequency Distribution: Arrangement of ungrouped data in the form of group is called Frequency Distribution of data.

Algorithm:

- First step is to find the “range” of the data values which can be partitioned into class interval. Use command `range` which returns a vector containing the minimum and maximum of all the given arguments.
- Create a sequence starting from lower limit to upper limit of given set of observations at an interval of width specified in the given problem.
- Now we need to convert Numeric to Factor using a command `cut`
- Now we can compute the absolute frequency of given data in each width interval with the `table()`
- Use the `cbind()` function to print the frequency distribution in column format.
- To compute the relative frequency distribution of given data in each width interval with `table()` by `length()`.

2.3. Cumulative Distribution Function (CDF) for data: It gives us an idea about the cumulative frequencies up to a certain point of the given data using `cumsum()`.

R Code:

```
>x=c(data vector)
```

```
>range(x) #returns a vector containing the minimum and maximum of all the given arguments.
```

```
>breaks = seq(minimum value, maximum value, by=specified width)
```

```
#Sequence at interval of specified width.
```

```
>C_I=cut(x, breaks, right = FALSE) #divides the range of data vector into intervals and codes
the values in data vector according to which interval they fall
>Freq=table(C_I) # gives the absolute frequency of the variable of the data file which generates
the frequency distribution of the data on variable
>cbind(Freq) # print the frequency distribution in column format
>Rel_Freq=table(C_I)/length(C_I) #gives the relative frequency of the variable of the data file
which generates the frequency distribution of the data on variable
>cbind(Rel_Freq) # print the relative frequency distribution in column format.
```

R Code for Cumulative Distribution function:

```
>Cum_Freq=cumsum(Freq) #returns a vector whose elements are the cumulative sums of the
elements of the absolute frequency distribution.
>cbind(Cum_Freq) # print the cumulative frequency distribution in column format
```

Note: Suppose if we want to display the total Frequency distribution table consisting of Absolute Frequency, Relative Frequency and Cumulative Frequency in column Format then:

```
> cbind(Freq, Rel_Freq, Cum_Freq)
```

1. Example:

Following are the 50 test scores for male students in an examination.

79, 70, 85, 64, 41, 69, 66, 28, 89, 47, 88, 72, 21, 81, 75, 72, 55, 35, 56, 80, 68, 52, 49, 35, 20, 30, 23, 48, 90, 24, 58, 65, 29, 50, 67, 37, 42, 36, 86, 63, 78, 84, 60, 22, 57, 61, 32, 27, 73, 31.

- Write R code to construct frequency distribution table of width 10.
- And also find relative frequency distribution and cumulative frequency distribution for the male students.
- Represent the above (a) and (b) in column format.

R code with Output:

2. Example:

Following are the scores of 250 students in a test with minimum and maximum marks being 0 and 100 respectively. The scores are stored in a data vector named **scores**.

21.08, 13.76, 75.89, 68.41, 51.36, 21.58, 13.85, 82.75, 71.88, 57.93, 6.68, 24.99, 10.24, 24.60, 95.14, 99.86, 42.72, 3.66, 66.89, 96.60, 30.08, 49.50, 16.84, 89.14, 14.97, 66.86, 38.32, 71.90, 9.50, 66.92, 58.83, 38.06, 79.95, 95.30, 97.51, 49.67, 50.54, 93.64, 74.30, 76.82, 63.32, 36.54, 90.87, 72.37, 61.98, 89.44, 98.19, 27.07, 63.97, 55.35, 92.40, 32.97, 61.37, 34.40, 42.48, 77.67, 99.31, 54.64, 69.46, 49.88, 86.76, 31.95, 4.55, 36.77, 36.91, 56.45, 54.44, 76.53, 15.97, 18.22, 41.34, 50.17, 18.82, 96.60, 93.58, 8.85, 30.27, 44.18, 26.85, 54.82, 64.24, 46.20, 3.00, 96.14, 10.01, 75.64, 93.18, 93.32, 15.88, 84.66, 37.91, 97.11, 66.32, 87.91, 85.58, 64.32, 16.11, 37.22, 42.69, 89.64, 89.96, 48.29, 27.71, 89.71, 23.08, 53.02, 4.22, 18.84, 17.41, 73.05, 33.26, 71.83, 8.28, 69.65, 51.27, 75.61, 63.67, 8.24, 1.15, 83.05, 23.08, 26.37, 19.74, 34.79, 53.58, 66.88, 99.58, 63.30, 60.82, 43.71, 39.76, 54.97, 13.55, 81.42, 26.04, 2.18, 69.18, 36.76, 67.77, 40.94, 37.88, 6.44, 24.03, 31.21, 44.02, 50.53, 1.29, 78.89, 78.18, 41.64, 13.12, 60.07, 37.38, 7.18, 38.98, 75.08, 17.65, 11.13, 19.43, 44.85, 63.36, 68.78, 91.34, 71.68, 90.47, 82.64, 15.24, 24.96, 14.07, 1.79, 35.54, 62.56, 76.10, 7.22, 96.00, 49.04, 93.72, 41.35, 6.35, 31.58, 14.91, 46.10, 48.83, 25.15, 15.98, 18.06, 54.47, 88.50, 87.06, 16.53, 47.53, 46.51, 68.91, 28.35, 68.89, 93.41, 21.07, 25.42, 93.02, 94.64, 93.94, 9.75, 89.52, 1.20, 34.41, 65.48, 58.09, 32.27, 55.53, 22.79, 5.41, 87.13, 52.83, 14.59, 17.07, 29.37, 94.04, 24.53, 62.59, 58.07, 83.71, 86.55, 23.05, 34.52, 35.22, 50.84, 71.60, 27.66, 5.29, 72.83, 2.01, 5.79, 72.59, 94.20, 1.20, 91.42, 34.59, 68.72, 6.51, 29.91, 40.08, 45.36, 16.20, 1.91, 22.10, 86.63, 44.83, 6.13, 93.60, 73.23.

- Write R code to construct frequency distribution table of width 15.
- And also find relative frequency distribution and cumulative frequency distribution for the male students.
- Represent the above (a) and (b) in column format.

R code with Output:

Viva Questions

1. Write R Code to represent the values in column format.
2. Define range
3. Write R code to find the relative frequencies

3. Acceptance/Rejection Sampling in R

Algorithm

1. Find a r.v. Y with density “ g ” satisfying $f(t)/g(t) \leq c$, for all t such that $f(t) > 0$.
2. Generate a random y from the dist. with density g
3. Generate a random u from the $U(0, 1)$ dist.
4. If $u < f(y)/(cg(y))$ accept y and deliver $x = y$; otherwise reject y and repeat step 2-4.

R code:

1. $U[0,1]$ vs $U[0,1]$

```
>install.packages("AR")
>library(AR)
> simulation = AR.Sim( n = 10,
  f_X = function(y){dunif(y,min=0,max=1)},
  Y.dist = "unif",
  Y.dist.par = c(0,1),
  Rej.Num = TRUE,
  Rej.Rate = TRUE,
  Acc.Rate = FALSE)
>simulation
```

2. Uniform (0,1) VS Standard Normal (0,1)

```
simulation = AR.Sim( n = 10,
  f_X = function(y){dunif(y,min=0,max=1)},
  Y.dist = "norm",
  Y.dist.par = c(0,1),
  Rej.Num = TRUE,
  Rej.Rate = TRUE,
  Acc.Rate = FALSE)
>simulation
```

3. Uniform (0,1) VS Normal (mean,var)

```
simulation = AR.Sim( n = 10,
  f_X = function(y){dunif(y,min=0,max=1)},
  Y.dist = "norm",
  Y.dist.par = c(mean,s.d),
  Rej.Num = TRUE,
  Rej.Rate = TRUE,
  Acc.Rate = FALSE)
>simulation
```

1. Example: Write R Program to simulate the data for $n=8$ using Acceptance-Rejection Algorithm for Uniform distribution in the interval $(0,1)$ and Normal Distribution with parameters mean 10 and variance 36 and Comment on your result.

R Code with Output:

2. Example: Write R Program to generate 8 observations using Acceptance-Rejection Algorithm for Uniform distribution in the interval (0,1) and Standard Normal Distribution and Comment on your result.

R Code with Output:

3. Example: Write R Program to generate 5 observations using Acceptance-Rejection Algorithm for Uniform distribution in the interval (0,1) and Uniform distribution in the interval (0,1) and Comment on your result.

R Code with Output:

ViVa Questions

1. Which package is used to generate random numbers in 'R' language
2. The mean and Variance of Poisson distribution is -----
3. The basic technique used to generate observations from a distribution is

4. Maximum likelihood and Method of moment's estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R

4.1 Maximum likelihood Estimation & Method of Moments estimation:

R Code:

```
> x=c(data vector)
> n=length(x)                                #No. of observations
> phat=mean(x)/n                             #MLE for Binomial distribution
where n represents no.of independent trails in Binomial distribution
> lambdahat=mean(x)                          #MLE for Poisson distribution
> muhat=mean(x)                              #MLE of Mean for Normal distribution
> sigmahat=((n - 1)/n)*var(x)                #MLE of variance for Normal distribution
```

Note: If the sample data is not given then generate the sample using random numbers of a particular distribution and then estimate the corresponding parameters.

R Code to generate the random numbers from Binomial, Poisson and Normal distribution

rbinom(no of random numbers to be generate, n, p)

rpois(n,lambda)

rnorm(n, mean, sd)

1. Example: Sample (9,2,5,3,1,5,2,4,2,1,3,5,6,1,1) is drawn from Binomial distribution. Use R program to find the maximum likelihood estimate of the parameter p and estimate the mean and variance.

R Code with Output:

2. Example: Let X follows Poisson distribution with parameter 12. Use R program to generate 10 samples from Poisson distribution and hence find the maximum likelihood estimate of the parameter and estimate the mean and variance.

R Code with Output:

3. Example: Sample (15, 9, 3, 7, 15, 8, 10, 3, 12, 5, 15, 8, 10, 3, 15, 8, 10, 3, 15, 8, 10, 3, 15, 7, 14, 10, 10, 14, 10, 9, 3, 3, 10, 15, 8, 10, 3, 15, 8, 10, 3, 12, 5, 15, 8, 10, 3, 15, 8, 10, 3) is drawn from Normal distribution. Use R program to estimate the mean and variance using method of moments.

R Code with Output:

4. Example: Sample (9, 3, 7, 12, 5, 15, 15, 8, 10, 3, 15, 7, 14, 10, 10, 14, 10, 9, 3, 3, 10) is drawn from Poisson distribution. Use R program to estimate the parameter, mean and variance of Poisson using Method of Moments.

R Code with Output:

4.2 Testing of Hypothesis (Large Samples)

> **Compute one sample proportion of z-test (Test for single proportion)**

R Code:

```
prop.test(x, n, p, alternative = "two.sided")
```

```
prop.test(x, n, p, alternative = "less")
```

```
prop.test(x, n, p, alternative = "greater")
```

x: the number of successes

n: the total number of trials

p: the probability to test against.

1. Example: A manufacture of spark plugs claims that 3% of items supplied by him defective. A random sample of 500 plugs is found to have 20 defectives items. test the claim of manufacture.

R Code with Output:

2. Example: We have a population of mice containing half male and half female ($p = 0.5 = 50\%$). Some of these mice ($n = 160$) have developed a spontaneous cancer, including 95 male and 65 female. Test whether the cancer affects more male than female?

R Code with Output:

> **Compute two-proportions of z-test (Test for difference of proportions)**

R Code:

```
prop.test(x=c(.....), n=c(.....), p = NULL, alternative = "..... ")
```

1. Example: In a large city A, consists of 900 school children, 20 had defective eyesight. In other large city B of a random sample 600 children, 15 had the same defect. Is this difference between the two proportions significant?

R Code with Output:

2. Example: In a company 500 people are having lung cancer and 500 people are healthy individuals. The number of smokers in each group is 490 and 400. Test whether the proportions of smokers are the same in the two groups of individuals?

R Code with Output:

C. One sample Z test in R (Test for single mean)

R Code:

```
install.packages("BSDA")    #BSDA: Basic Statistics and Data Analysis
library(BSDA)
z.test(datavector, mu=population mean, sigma.x=....., alternative= ".....")
```

1. Example: The price of a popular tennis racket at a national chain store is \$179. Portia bought ten of the same racket at an online auction site for the following prices:

155,179,175,170,175,173,172,171,170,169,168,164,165,162,163,172,176,174,168,165,170,172, 168,162,175,161,160,158,156,171,170. If the auction prices of rackets are normally distributed, Use R code to determine whether there is sufficient evidence in the sample, at the 5% level of significance, to conclude that the average price of the racket is less than \$179 with standard deviation 6.1 if purchased at an online auction.

R Code with Output:

d.Two Sample Z-Test in R (Test for difference of means)

R Code:

```
z.test(datavector1,datavector2,      alternative='.....',      mu=0,      sigma.x= .....,
sigma.y= .....) )
```

1. Example: Suppose the IQ levels among individuals in two different cities are known to be normally distributed each with population standard deviations of 15 and 12 respectively. A scientist wants to test whether the mean IQ level between individuals in city A and city B are different, so she selects a [simple random sample](#) of 32 from city A and 36 samples from city B and records their IQ levels.

[Hint: First generate random samples using rnorm]

R Code with Output:

Viva Questions:

1. Define Null Hypothesis and Alternative Hypothesis.
2. Write the test statistic to test the difference of two proportions.
3. Write the test-statistic to test for single mean in large samples.

5. Testing of hypothesis based on t-test, chi-square test and F-test. Confidence interval estimation in R (Small Samples)

5.1 t test

i) t-test for single mean

R Code:

t.test(data vector, mu = population mean, alternative= “two.sided”, conf.level=0.98)

t.test(data vector, mu = population mean, alternative= “less”)

t.test(data vector, mu = population mean, alternative= “greater”)

Note: Conf.level is an additional argument when one wants to construct 90% or 98% or 99%, we need to write. Otherwise by default it will construct 95% confidence interval without writing this additional argument.

1. Example: A random sample of 10 envelopes is taken from the letter box of a post office and their weights in grams are found to be 13.1, 14.5, 12.1, 11.9, 12.4, 12.3, 11.5, 11.6, 12.1 and 12.4. Does this sample indicate at 1% level that the average weight of envelopes received at their post office is 12.35 grams? Also construct 99% confidence interval for population mean.

R Code with Output:

2. Example: Test the hypothesis that the average content of containers of a particular lubricant is 10 liters if the content of a random sample of 10 containers are:

10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, 9.8

Use 0.01 l.o.s and assume that the distribution of contents is normal. Also construct 95% confidence interval for population mean.

R Code with Output:

ii) Two sample t test (Test for difference of means_independent samples)**R Code:****t.test(data vector 1, data vector 2, mu = 0, alternative= “.....”)**

1. **Example:** Two Horses A and B were tested according to the time (seconds) in a particular track with the following results.

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	-

Test whether the two horses have same running capacity? Also construct 95% confidence interval for population mean.

R Code with Output:

iii) Paired t test (Test for difference of means_dependent samples)

R Code:

t.test(after, before, paired = TRUE, alternative = ".....")

1. Example: Memory capacity of 10 students was tested before and after training. Test whether the training was effective or not from the following scores. Also construct 99% confidence interval for population mean.

Before training	12	14	11	8	7	10	3	0	5	6
After training	15	16	10	7	5	12	10	2	3	8

R Code with Output:

5.2 Chi-square test

R Code:

x=matrix(c(data),ncol=,nrow=.....,byrow=TRUE)

or

x =data.frame(col1 = c (.....), col2 = c(.....),.....)

x

chisq.test(x)

1. Example: 1000 students at college level were graded according to their I.Q and the economic conditions of their home. Write R code to test whether there is any association between condition at home and I.Q.

	I.Q		
Economic	High	Low	Total
Rich	460	140	600
Poor	240	160	400
Total	700	300	1000

R Code with Output:

Example: In an industry, 1000 workers, employed for a specific job, were classified according to their performance and training received to test independence of a specific training and performance. The data is summarized as follows.

Training	Performance	
	Good	Not Good
Trained	230	148
Untrained	151	471

Write R code to make use of Chi-square test for independence at 5 % level of significance and write your conclusion.

R Code with Output:

5.3 F-test. (Test for equality of variances)

R Code:

```
var.test(sample1,sample2,alternative="....")
```

Note: While calculating F ratio, F value should be greater than 1. If not interchange the samples.

1. Example: Two Horses A and B were tested according to the time (seconds) in a particular track with the following results.

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	29	-

Test at 5% level whether the populations have the same variance.

R Code with Output:

2. Example: The following are the values in thousands of an inch obtained by 2 engineers in 8 successive measurements with the same micrometer:

Engineer A: 503,505,497,505,419,493,510,501

Engineer B: 502,497,492,490,495,497,596,498

Is one engineer more consistent than the other?

R Code with Output:

6. Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi collinearity in R

6.1 Fitting of Multiple Linear Regression

R Code:

MLR<- lm(Dependent variable ~ sum of independent variables)

➤ Testing MLR

R Code:

summary(MLR)

6.2. Outlier analysis

R Code:

plot(MLR)

6.3. Residual analysis

R Code

Residuals=resid(MLR)

Predicted=predict(MLR)

Residuals

Predicted

plot(Residuals,Predicted)

6.4. Test for normality

R Code:

plot(MLR,2)

2 indicates second plot i.e Normal Q-Q plot

shapiro.test(Residuals)

to test the Residuals follows normality or not

6.5. Multi Collinearity

R Code:

install.packages("caTools")

For Linear

regression

install.packages('car')

To check

multicollinearity

install.packages("quantmod")

install.packages("MASS")

install.packages("corrplot")

plot

correlation plot

library(caTools)

library(car)

library(quantmod)

library(MASS)

library(corrplot)

vif(MLR) **#Variance Inflation Factor**

(vif) specifies whether Multicollinearity

present in the variables or not

if vif value is >10 then there exist

Multicollinearity

1. Example:

The diameter of a tree at its breast height is influenced by many factors. The below table relate the diameter of a particular type of eucalyptus tree to its age, average rainfall at its site, site's elevation, and the wood's mean specific gravity.

S.No.	Age (Years)	Elevation (1000 ft)	Rainfall (inches)	Specific Gravity	Diameter at Breast Height (inches)
1	35	1.5	110	0.63	18.1
2	36	1.2	115	0.59	19.6
3	34	2.1	80	0.56	16.6
4	35	2.3	89	0.55	16.4
5	36	2.5	78	0.54	16.9
6	32	1.6	54	0.59	17.0
7	35	1.5	86	0.56	20.0
8	38	1.4	110	0.46	16.6
9	39	2.3	118	0.63	16.2
10	34	2.5	119	0.60	18.5
11	32	2.6	200	0.63	18.7
12	31	1.2	240	0.58	19.4
13	35	1.3	210	0.55	17.6
14	36	1.8	110	0.57	18.3
15	32	1.9	118	0.62	18.8

- Fit a Multiple linear regression model and Test the Model
 - Identify the Outliers in the data
 - Check for Multicollinearity for the variables
 - Compute the Residuals
 - Test for Normality
- a) R Code with Output:**

b) R Code with Output:

c) R Code with Output:

```
install.packages("caTools") # For Linear regression  
install.packages('car') # To check multicollinearity  
install.packages("quantmod")  
install.packages("MASS")  
install.packages("corrplot") # plot correlation plot
```

d) R Code with Output:

e) R Code with Output:

2. Example: The data given below consists of 10 observations on the response (Y) effected by the independent variables X1, X2, X3, X4. Use R code:

- Fit a Multiple linear regression model
- Identify the Outliers in the data
- Test for Normality

Y	21.0	22.0	21.0	23.0	21.2	23.2	21.5	22.0	22.3	22.5
X1	2.3	2.5	2.6	2.5	2.4	2.1	2.0	2.2	2.3	2.9
X2	3.1	3.2	3.3	3.4	3.5	3.6	3.8	3.9	3.2	3.3
X3	4.1	4.2	4.3	4.4	4.5	4.6	4.6	4.8	4.9	4.2
X4	5	6	7	8	9	1	5	4	2	6

a) R Code with Output:

b) R Code with Output:

C) R Code with Output:

3. Fit Multiple Linear regression to Marketing data which is available in “R”.

- a) Identify the Outliers in the data
- b) Test for Normality

Note: This data can be downloaded using the package "datarium"

R Code: data("marketing", package = "datarium")

head(marketing, 4) #Displays first 4 values in the data

head(marketing, 4)

tail(marketing) #Displays last 6 values in the data

View(marketing) #Displays the total data

R Code with Output:

a) R Code with Output:

b) R Code with Output:

Viva Questions

1. Define Linear Regression.
2. Define Residual.
3. Define Multicollinearity and when Multicollinearity exist in linear regression.

7. Logistic Regression in R

R Code:

```
Logistic_Reg = glm(Dependent variable ~ sum of Independent variables,
family="binomial")
```

```
summary(Logistic_Reg)
```

1. Example:

Consider data published on $n = 27$ leukemia patients. The data has a response variable of whether leukemia remission occurred (REMISS), which is given by a 1. The predictor variables are cellularity of the marrow clot section (CELL), smear differential percentage of blasts (SMEAR), percentage of absolute marrow leukemia cell infiltrate (INFIL), percentage labeling index of the bone marrow leukemia cells (LI), absolute number of blasts in the peripheral blood (BLAST), and the highest temperature prior to start of treatment (TEMP).

REMISS	CELL	SMEAR	INFIL	LI	BLAST	TEMP
1	0.8	0.83	0.66	1.9	1.1	1
1	0.9	0.36	0.32	1.4	0.74	0.99
0	0.8	0.88	0.7	0.8	0.18	0.98
0	1	0.87	0.87	0.7	1.05	0.99
1	0.9	0.75	0.68	1.3	0.52	0.98
0	1	0.65	0.65	0.6	0.52	0.98
1	0.95	0.97	0.92	1	1.23	0.99
0	0.95	0.87	0.83	1.9	1.35	1.02
0	1	0.45	0.45	0.8	0.32	1
0	0.95	0.36	0.34	0.5	0	1.04
0	0.85	0.39	0.33	0.7	0.28	0.99
0	0.7	0.76	0.53	1.2	0.15	0.98
0	0.8	0.46	0.37	0.4	0.38	1.01
0	0.2	0.39	0.08	0.8	0.11	0.99
0	1	0.9	0.9	1.1	1.04	0.99
1	1	0.84	0.84	1.9	2.06	1.02
0	0.65	0.42	0.27	0.5	0.11	1.01
0	1	0.75	0.75	1	1.32	1
0	0.5	0.44	0.22	0.6	0.11	0.99
1	1	0.63	0.63	1.1	1.07	0.99
0	1	0.33	0.33	0.4	0.18	1.01
0	0.9	0.93	0.84	0.6	1.59	1.02
1	1	0.58	0.58	1	0.53	1
0	0.95	0.32	0.3	1.6	0.89	0.99
1	1	0.6	0.6	1.7	0.96	0.99
1	1	0.69	0.69	0.9	0.4	0.99
0	1	0.73	0.73	0.7	0.4	0.99

R Code with Output:

2. Fit Logistic Regression to Student admit data and write the appropriate results

Note: The data can be download using the following code

```
install.packages("packagename")
```

```
mydata <- read.csv("https://stats.idre.ucla.edu/stat/data/binary.csv")
```

R Code with Output:

Viva Questions:

1. What kind of Machine Learning is Logistic Regression?
2. Define Binary Response Variable?
3. State any two differences of Linear Regression and Logistic Regression?

8. Classification with SVM in R

R Code:

```
install.packages("e1071")
library(e1071)
x = matrix(rnorm(32), 16, 2)           #Generating the data
y = rep(c(-1, 1), c(8, 8))             #Assigning +1 and -1 to the above data
x[y == 1,] = x[y == 1,] + 1           #bias the x variable when y =1
dat = data.frame(x, y = as.factor(y))  #representing the above data in table form using dataframe
svm = svm(y ~ ., data = dat, kernel = "linear", cost=10)
summary(svm)
```

1. Example: Write R code to find the number of support vectors for classifying the data. Data as follows: 60 observations, in two columns and also get the SVM classification plot.

R Code with Output:

2. Example: Write R code to find the number of support vectors for classifying the data. Data as follows: 100 observations, in two columns and also get the SVM classification plot.

R Code with Output:

Viva Questions:

- 1 Define SVM?
2. Define Hyperplane?
3. Define support vectors?

9. Dimensionality reduction with PCA in R and K-means Clustering in R

9.1 Principal Component Analysis (PCA) in R

R Code:

Store the **Data** in Excel and Export the **Data**

PCA=prcomp(Data, scale=T)

summary(PCA)

1. Example: Obtain the percentage of principal components to the following data and interpret the results

A	38	35	36	38	35	30	38	30	26	32	35	36	31	38	35	30	29
B	61	61	62	65	60	61	61	62	60	58	56	57	58	65	60	62	52
C	33	33	65	67	59	33	33	56	60	62	61	60	52	67	59	56	59
D	32	32	27	34	43	32	32	44	48	52	33	39	42	34	43	44	56
E	50	50	31	33	36	50	50	35	31	30	32	29	30	33	36	35	31
F	38	35	54	57	70	30	38	72	77	49	50	51	56	57	70	72	52
G	65	60	65	74	60	62	65	62	61	68	58	62	61	74	60	62	63

R Code with Output:

2. Example: Obtain the percentage of principal components to the following data and interpret the results

X1	44	33	33	32	34	31	33	30	34	34	33	36	33	34	37
X2	1.3	2.2	2.2	2.6	2	1.8	2.2	3.6	1.6	1.5	2.2	1.7	2.2	1.3	2.6
X3	250	115	75	85	100	75	85	75	225	250	255	175	75	85	90
X4	0.63	0.59	0.56	0.55	0.54	0.59	0.56	0.46	0.63	0.6	0.63	0.58	0.55	0.57	0.62

R Code with Output:

9.2 K-means Clustering in R

Algorithm:

1. Install appropriate package
2. Store the **Data** in Excel and Export the **Data**
3. Choose k value by any one of the three methods
4. Apply kmeans clustering to the given data.

R Code:

```
install.packages("factoextra") # To visualize the clusters
library(factoextra)
```

To choose k value: 3 methods are (Elbow, Silhouette, Gap)

```
fviz_nbclust(Data, kmeans, method = "wss", k.max = 8) #k.max argument is optional
```

or

```
fviz_nbclust(Data, kmeans, method = "silhouette", k.max = 8)
```

or

```
fviz_nbclust(Data, kmeans, method = "gap", k.max = 8)
```

#From the above 3 methods k value will be taken

```
clust=kmeans(Data, k value)
```

```
clust
```

```
fviz_cluster(clust, Data) #Displays cluster plot
```

1 Example: The following is the set of 10 data points. Cluster these points using R code in k means algorithm.

S.No.	1	2	3	4	5	6	7	8	9	10
X ₁	18	22	25	32	38	42	44	46	52	60
X ₂	12	18	25	28	32	38	40	43	45	50
X ₃	18	22	35	38	39	43	50	55	60	65

R Code with Output:

2 Example: Cluster these data points using R code in k means algorithm.

X1	73	63	65	60	62	61	71	72	73	63	59	60	71	72	65
X2	60	62	61	63	60	56	65	64	60	62	62	62	65	64	55
X3	48	45	45	45	47	41	41	42	48	45	41	43	41	42	42
X4	38	44	74	70	68	42	32	35	38	44	35	30	32	35	56
X5	70	77	52	67	65	66	65	60	70	77	62	61	65	60	52
X6	65	63	62	67	65	64	58	62	65	63	60	56	58	62	62

R Code with Output:

Viva Questions:

1. Define Cluster.
2. Define Principal Components.
3. What kind of Machine learning did PCA and Clustering techniques?

10. Kernel SVM in R

Kernel SVM:

R Code:

```
install.packages("e1071")
library(e1071)
x = matrix(rnorm(32), 16, 2)           #Generating the data
y = rep(c(-1, 1), c(8, 8))             #Assigning +1 and -1 to the above data
x[y == 1,] = x[y == 1,] + 1           #bias the x variable when y =1
dat = data.frame(x, y = as.factor(y))  #representing the above data in table form using dataframe
svm = svm(y ~ ., data = dat, kernel = "radial", gamma=1, cost=1)
summary(svm)
plot(svm, dat)
```

1 Example: Write R code to find the number of support vectors for classifying the data in radial basis kernel. Data as follows: 400 observations, in two columns and also get the SVM classification plot.

R Code with Output:

2 Example: Write R code to find the number of support vectors for classifying the data in radial basis kernel. Data as follows: 300 observations, in two columns and also get the SVM classification plot.

R Code with Output:

Viva Questions:

1. Define Kernel.
2. What is the use of Kernel

List of Additional programs

1. Generating Functions (Binomial, Poisson, Uniform, Normal and Exponential) using R programming

(i) Uniform Distribution

R Code:

```
runif(n, min, max)
```

1. **Example:** Generate 5 Uniform random numbers from 10 to 100.

R Code with Output:

2. **Example:** Generate 10 random samples from the Uniform Distribution $U(0,1)$

R Code with Output:

(ii) Normal Distribution

R Code:

```
rnorm(n, mean, sd)
```

1. **Example:** Generate four random numbers from $N(70, 10)$

R Code with Output:

2. **Example:** Generate 5 Random samples from the Normal distribution $N(25,4)$

R Code with Output:

(iii) Binomial Distribution

R Code:

```
rbinom(no of random numbers to be generate, n, p)
```

1. **Example:** Generate 6 Random samples from the Binomial distribution B (10, 0.5)

R Code with Output:

(iv) **Exponential Distribution**

R Code:

`rexp(n, theta )`

1. **Example:** Generate 5 Random samples from the Exponential distribution with parameter 10

R Code with Output:

2. Fit Simple linear regression to the given data

Hours: 1, 2, 4, 5, 5, 6, 6, 7, 8, 10, 11, 11, 12, 12, 14

Score: 64, 66, 76, 73, 74, 81, 83, 82, 80, 88, 84, 82, 91, 93, 89

R Code with output:

3. Multiple linear regression

Testing overall hypothesis and testing significance of individual variables, model selections and prediction in R programming is called Multiple linear regression.

R Code:

`lm(y ~ x1+x2+x3...,data)`

Example 1: Fit a multiple linear regression to the following data set.

S.No.	1	2	3	4	5	6	7	8	9	10	11	12
X1	70	77	72	77	74	80	82	74	72	77	90	82
X2	100	105	112	105	125	112	129	125	112	105	140	129
X3	125	110	118	110	122	123	121	122	118	110	128	121
X4	65	70	66	70	58	68	61	58	66	70	82	61
Y	77	73	68	76	73	77	80	73	72	74	66	78

R Code with Output:

Programme Outcomes

S N o.	CO	Name of the Programme	Outcome of the Programme	Mention any Real-time application where we can apply this method	Which one is more efficient
1	CO1	Introduction to R Programming	Perform built-in commands and operations with matrices in R-programming. Analyze the appropriate data type of variables and evaluate various Frequency Distributions in R - programming.	R is a open source Software and it can be easily accessed. It represents the whole set of data by a single value. and gives us the location of central points.	
		Introduction to descriptive statistics using R (Frequency Distribution and Cumulative Distribution Function)			
2	CO2	Acceptance/Rejection Sampling in R	Determine the mean and standard deviation of a sampling distribution using R.	Apply in Bayesian inference for complex models where direct sampling is difficult.	
3	CO3	Maximum likelihood and method of moments estimation. Testing of Hypothesis based on Z-test. Concept of p-value in R. Testing of hypothesis based on t-test, chi-square test and F-test. Confidence interval estimation in R	Develop a code for testing of hypothesis using Z-test, t-test, chi-square-test, F-test.	To test the significance of hypothesis based on the p-value	
4	CO4	Multiple linear regression: Outlier analysis, residual analysis, test for normality, multi-collinearity in R	Classify Linear and Logistic regression. Write Code for SVM and Kernel SVM methods.	To classify the objects into groups when the data is large.	
		Logistic Regression in R			
		Classification with SVM in R			
		Dimensionality reduction with PCA in R and K-means Clustering in R			
		Kernel SVM in R			



...striving towards perfection